SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

# AI-BASED DIGITAL LIFESTYLE AND WELLBEING ANALYSIS AMONG YOUNG ADULTS

**ADA23SLU07 – RESEARCH PRODUCTIVITY TOOLS**

**PROJECT REPORT**

***Submitted by***

***Nikilesh Gajan J R – E0324098***

***In partial fulfilment for the award of the degree of***

**BACHELOR OF TECHNOLOGY**

**in**

**COMPUTER SCIENCE AND ENGINEERING**

**Artificial Intelligence and Data Analytics**

**Sri Ramachandra Faculty of Engineering and Technology**

**Sri Ramachandra Institute of Higher Education and Research, Porur, Chennai -600116**

**October 2026**

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# BONAFIDE CERTIFICATE

This is certified to be the Bonafide work of the student **Nikilesh Gajan J R(E0324098)** in the Research Productivity Tools ADA23SLU07 , at Sri Ramachandra Faculty of Engineering and Technology, in partial fulfillment of Semester 5, Third Year B.Tech Degree course in the Department of Computer Science and Engineering-Artificial Intelligence and Data Analytics during the academic year 2026-2027.

**Signature of the Supervisor**

**Dr. Deena S**

**Assistant Professor,**

Department of Artificial Intelligence and Data Analytics,

Sri Ramachandra Faculty of Engineering and Technology,

SRIHER(DU), Porur, Chennai-600 116.

**Evaluation Date:**

**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

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I am grateful to all the members of the Sri Ramachandra Faculty of Engineering and Technology, my beloved parents and my friends for extending their support, which helped us to overcome obstacles in the study.

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## Experiment 1: Literature Search and Research Article Identification Using PubMed and Google Scholar

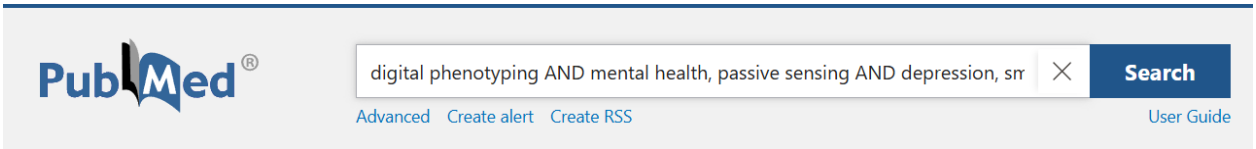
**Aim:** To perform a systematic literature search using PubMed and identify relevant research articles using appropriate keywords, filters and search strategies.

**Tools Used:** Google Scholar

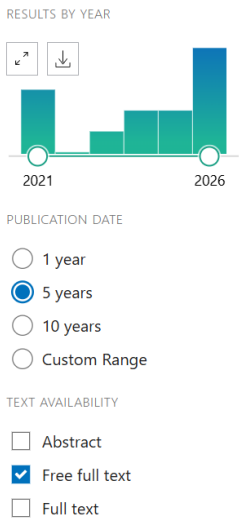
**Procedure:**

1. Open PubMed/ Google Scholar
2. Define the research topic and keywords.
3. Perform keyword-based searches.
4. Apply filters such as publication year, article type and availability.
5. Identify relevant research articles.
6. Record and evaluate the retrieved literature.

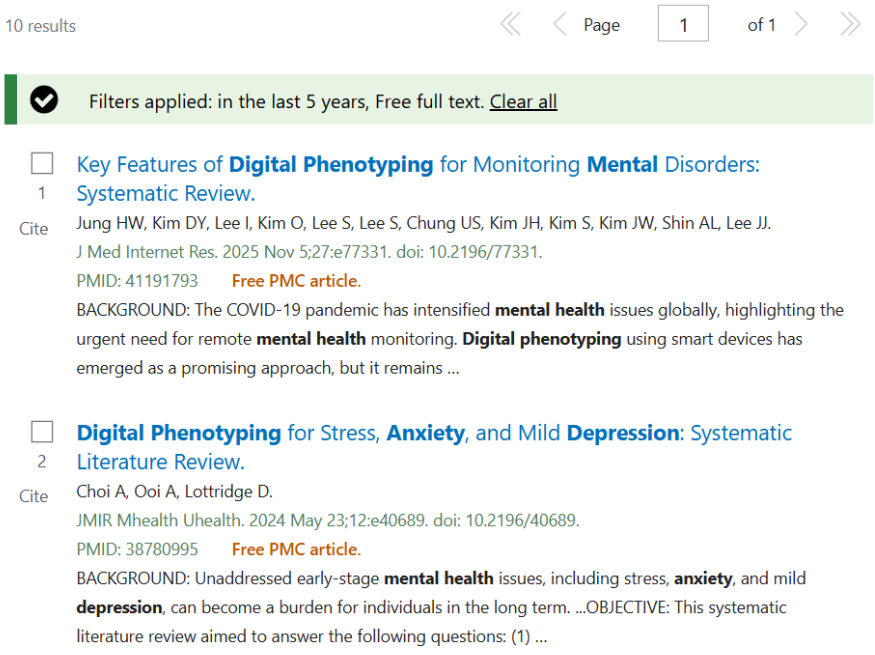
**Output:**



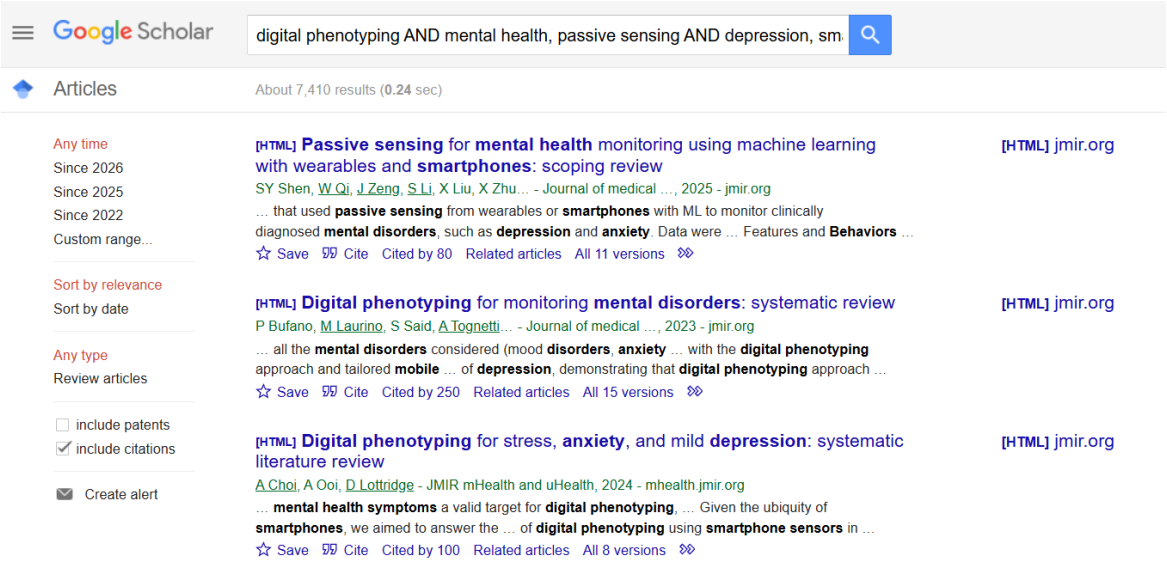

IMG 1.1: *Pubmed search bar*




IMG 1.2: *Pubmed Search filters*




IMG 1.3: *Pubmed Search Results*




IMG 1.4: *Google scholar search results*

**Result:** Relevant research articles were successfully identified and organized using appropriate literature-search techniques.

**Conclusion**: PubMed provides an effective platform for systematic identification of biomedical and healthcare research literature.

## Experiment 2: Comparative Literature Search Using Google Scholar, Connected Papers and Research Rabbit

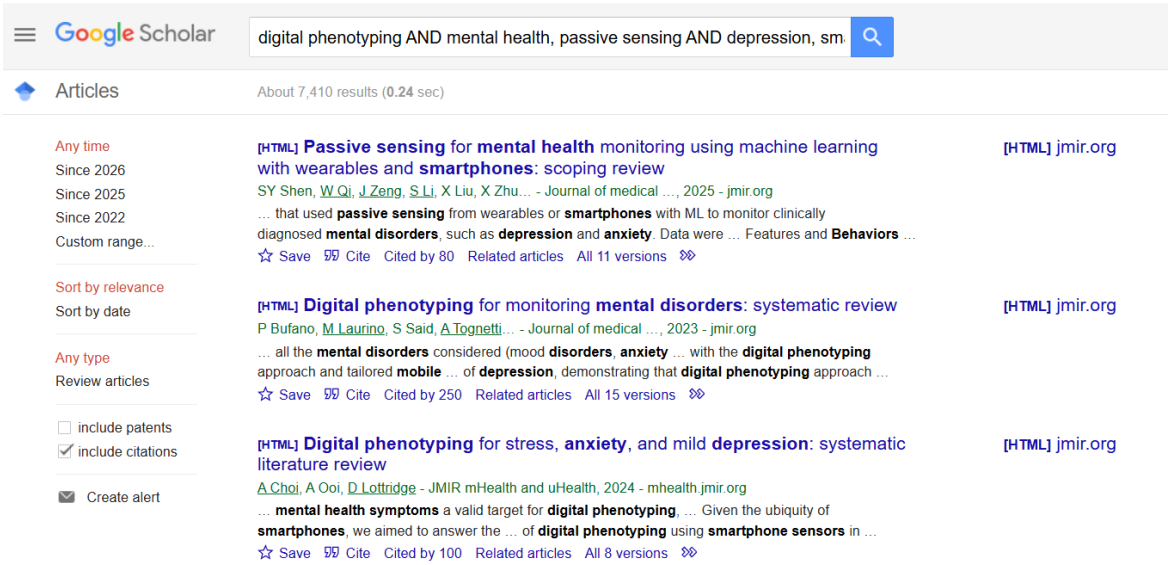
**Aim:** To perform a comparative literature search using Google Scholar, Connected Papers and Research Rabbit to identify, explore and organize relevant research articles.

**Tools Used:** Google Scholar, Connected Papers and Research Rabbit

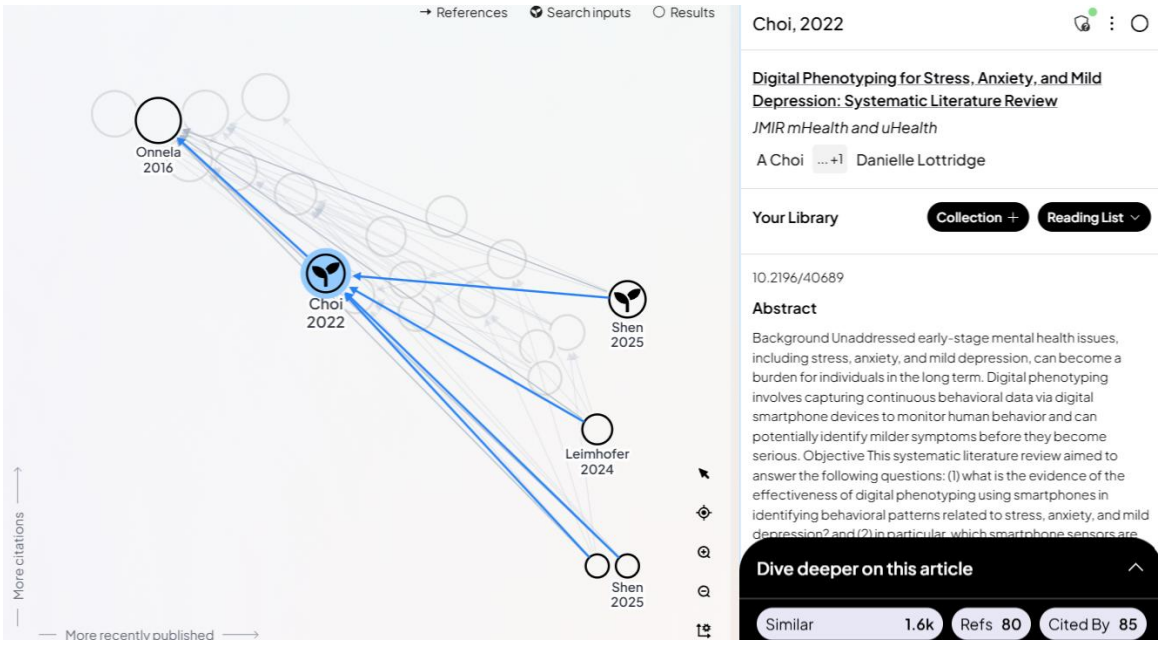
**Procedure:**

1. Open Google Scholar, Connected Papers and Research Rabbit.
2. Define the research topic and relevant keywords.
3. Search for research articles using appropriate keywords in Google Scholar.
4. Identify related research papers using Connected Papers.
5. Explore authors, papers and research connections using Research Rabbit.
6. Compare and record the relevant literature obtained from all three platforms.

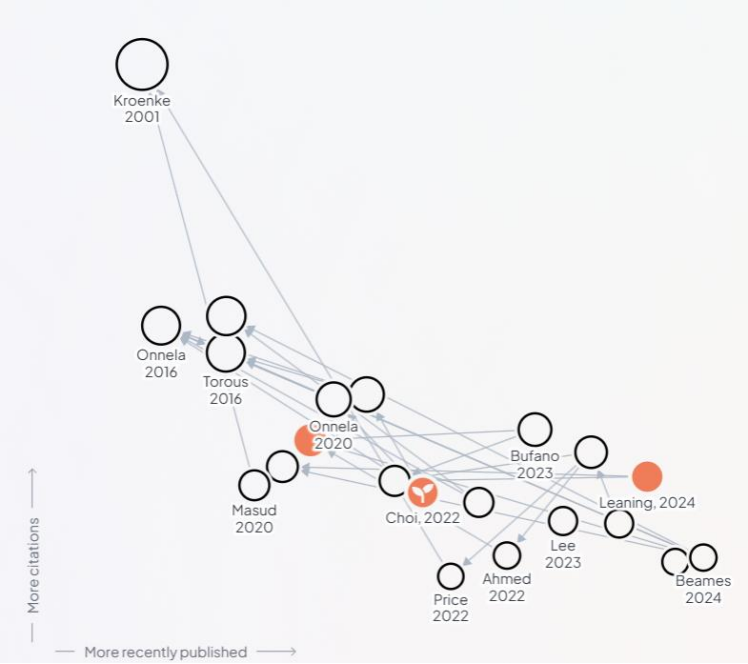
**Output:**




IMG 2.1: *Google scholar with filters, 7410 results*




IMG 2.2: *Research rabbit research for articles*




IMG 2.3: *Research rabbit articles displayed as a map*




IMG 2.4: *Connected papers articles research*

**Result:** Relevant research articles were successfully identified, explored and compared using multiple literature-search platforms.

**Conclusion:** Google Scholar, Connected Papers and Research Rabbit provide effective tools for searching, discovering and organizing related research literature.

## Experiment 3: Creation of Professional Research Diagrams and Flowcharts Using Draw.io

**Aim:** To create professional research diagrams and flowcharts using Draw.io for representing research processes and workflows.

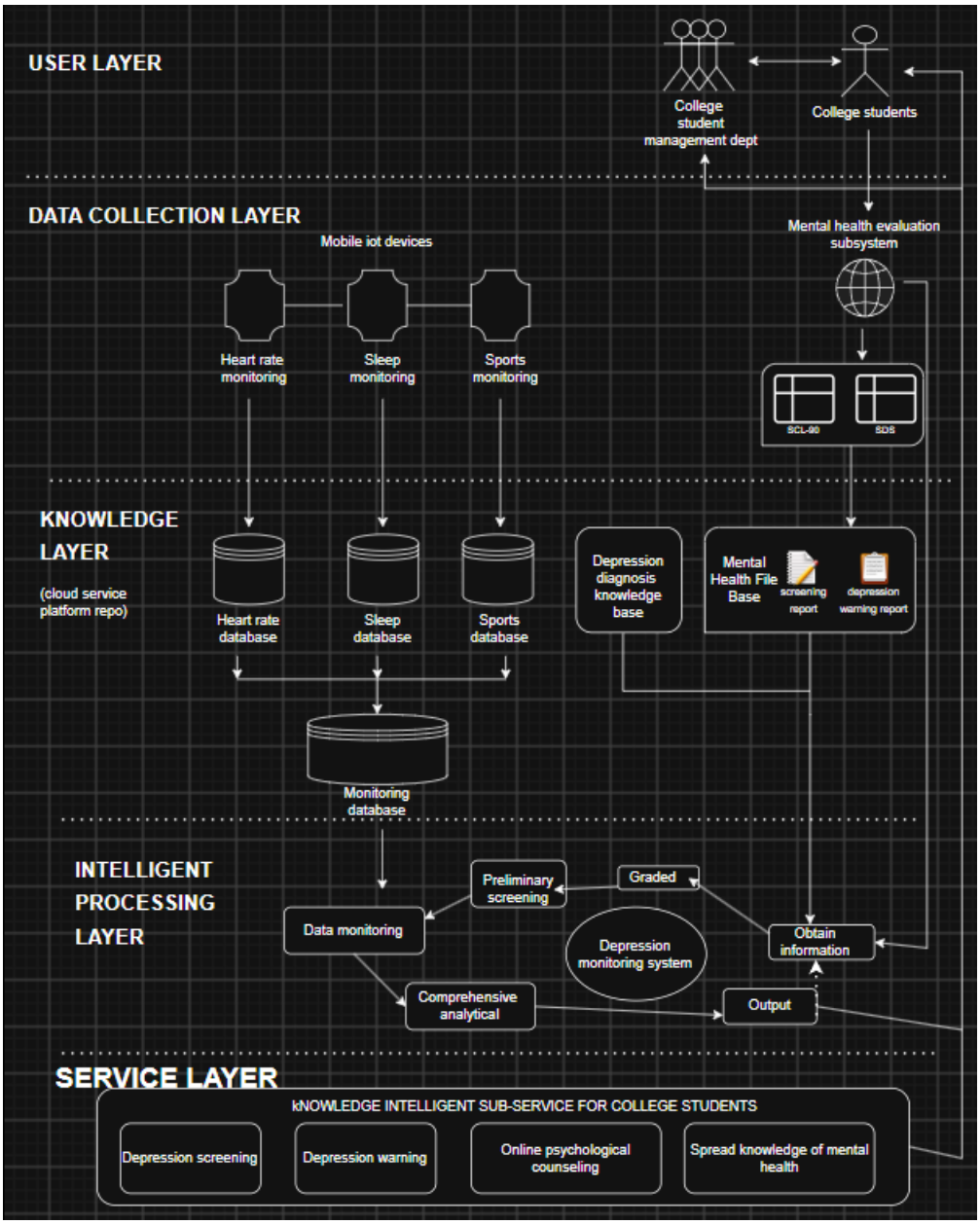
**Tools Used:** Draw.io

**Procedure:**

1. Open Draw.io.
2. Define the research process or workflow to be represented.
3. Select appropriate shapes, symbols and connectors.
4. Create the research diagram or flowchart using the selected elements.
5. Arrange and format the elements for clear and professional representation.
6. Save and export the completed research diagram.

**Output:** Insert your Draw.io diagram/flowchart screenshots here.

**Result:**




IMG3.1: *DRAW.IO image*

**Conclusion:** Draw.io provides an effective platform for creating clear, structured and professional research diagrams and flowcharts.

## Experiment 4: Data Visualization Using Tableau: Charts and Graphical Representations

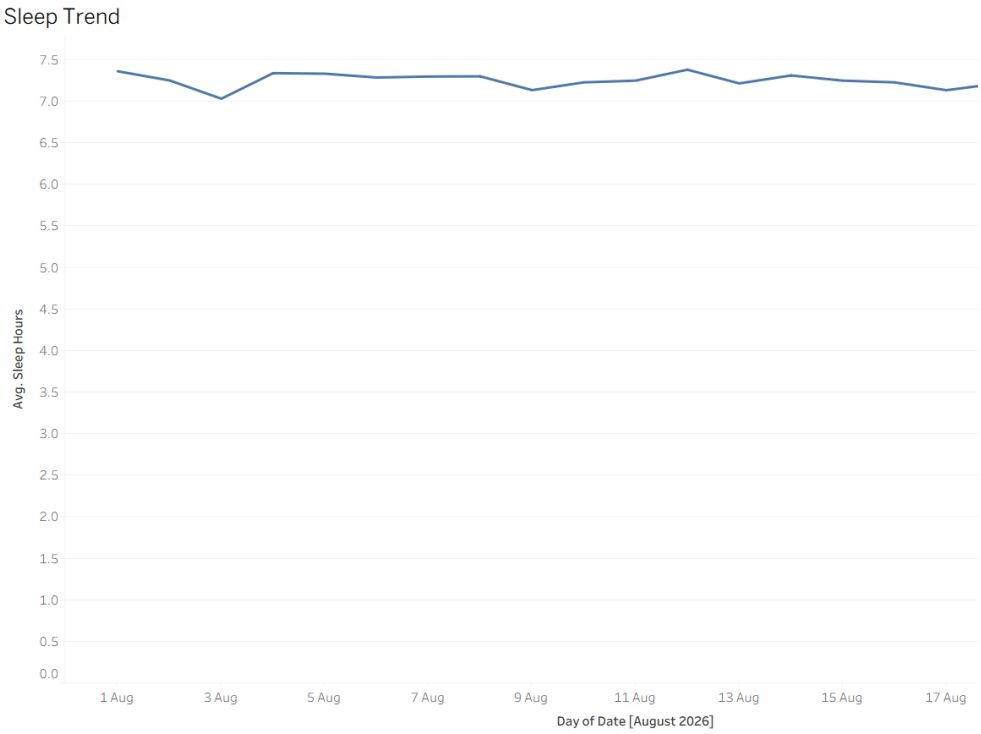
**Aim:** To visualize research data using Tableau and create appropriate charts and graphical representations for effective data interpretation.

**Tools Used:** Tableau

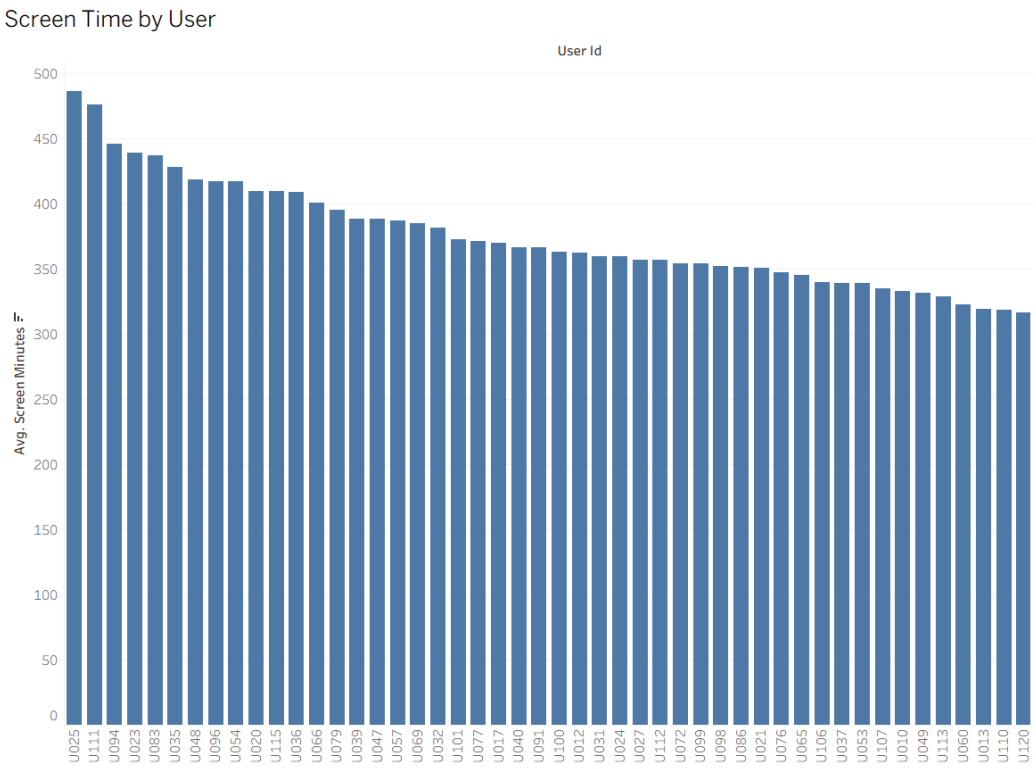
**Procedure:**

1. Open Tableau.
2. Import the required research dataset.
3. Examine the available fields and data types.
4. Select appropriate charts based on the data and analysis requirements.
5. Create graphical representations using dimensions and measures.
6. Format and organize the charts for clear visualization.

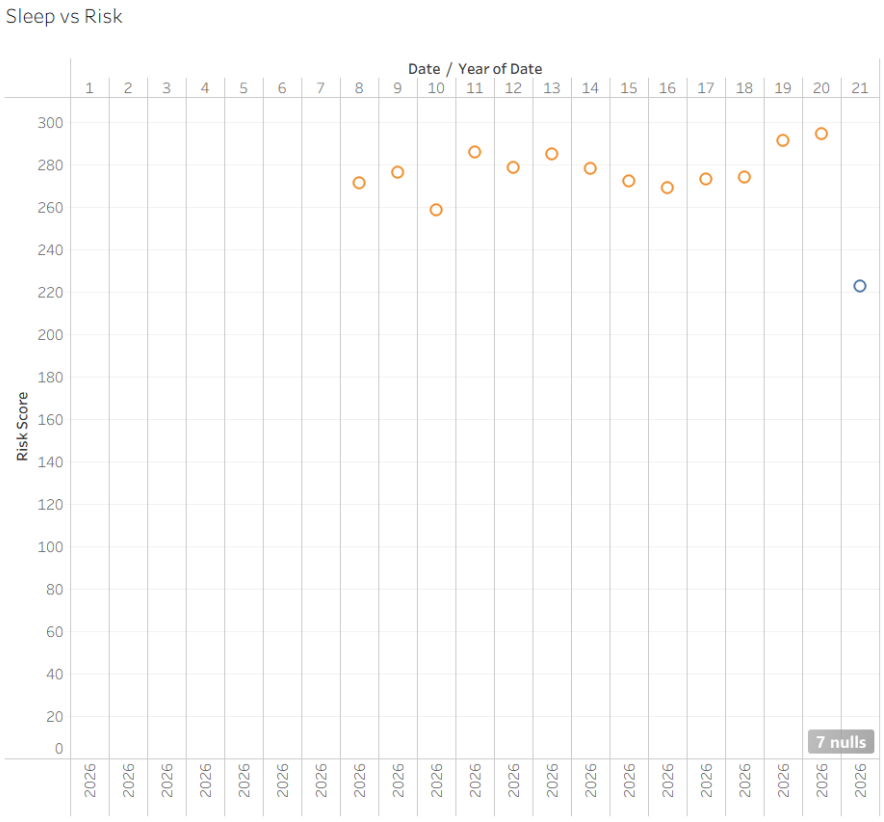
**Output:**



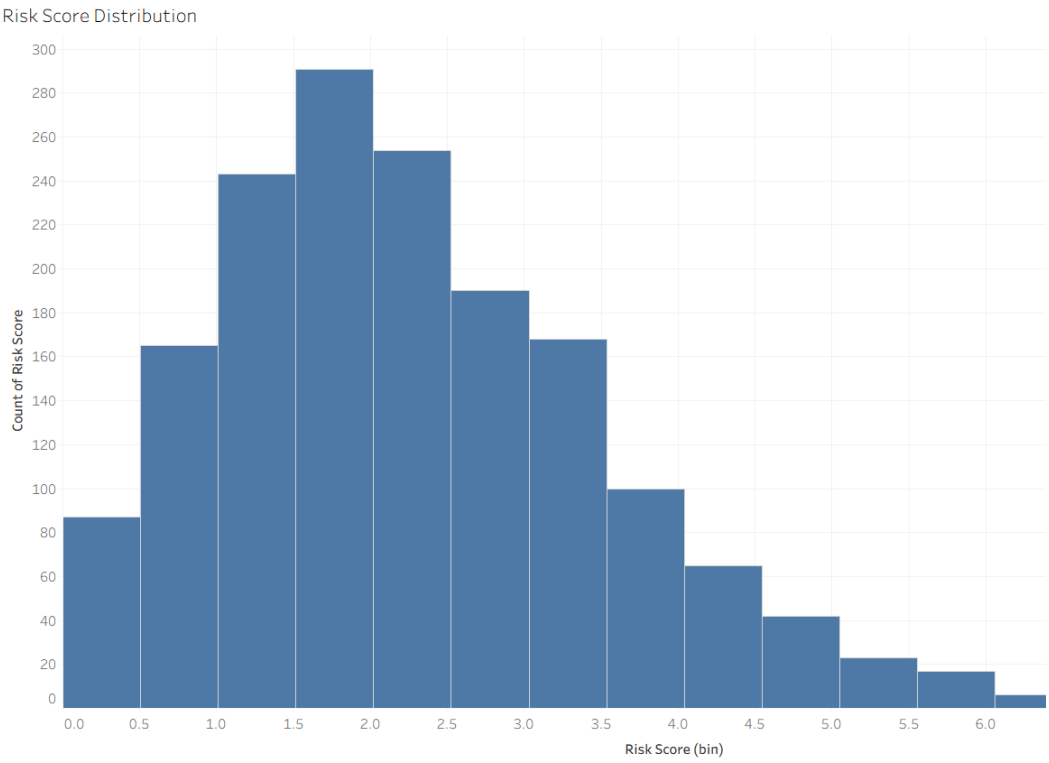

IMG 4.1: *Sleep trend - line chart*




IMG 4.2: *Screen time by user - bar chart*




IMG 4.3: *Sleep Hours - Scatter plot*




IMG 4.4: *Distribution of risk scores - Histogram*

**Result:** Research data was successfully visualized using appropriate charts and graphical representations in Tableau.

**Conclusion:** Tableau provides an effective platform for transforming research data into meaningful and easily interpretable visual representations.

## Experiment 5: Advanced Data Analysis and Report Generation Using Tableau

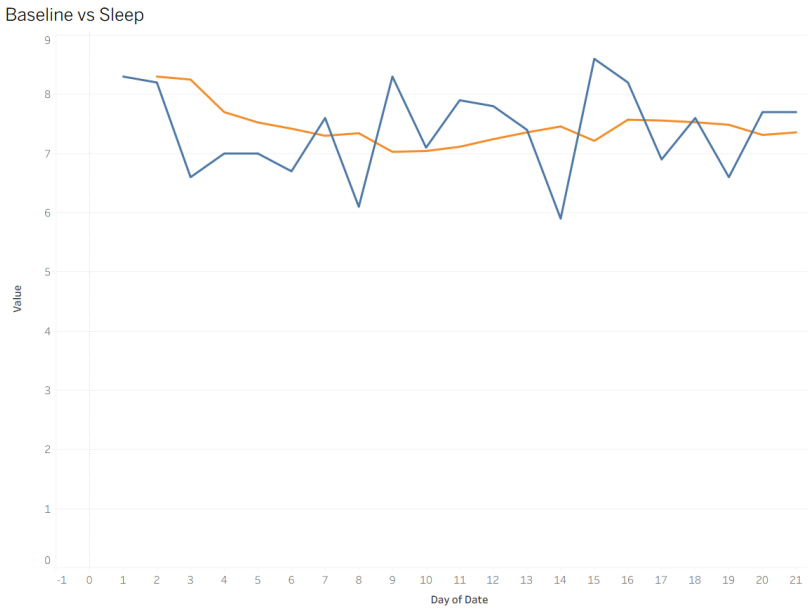
**Aim:** To perform advanced data analysis and generate meaningful analytical reports using Tableau.

**Tools Used:** Tableau

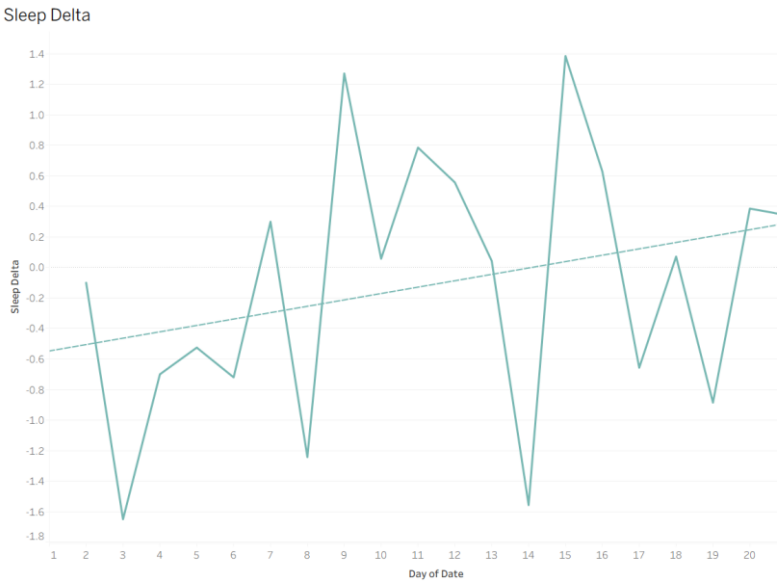
**Procedure:**

1. Open Tableau and import the required dataset.
2. Examine and prepare the data for analysis.
3. Apply appropriate filters, calculations and analytical techniques.
4. Create charts and visualizations based on the analysis requirements.
5. Analyze the visualized data to identify important patterns and trends.
6. Generate and format a comprehensive analytical report.

**Output:**




IMG 5.1: *Baseline vs sleep line chart*




IMG 5.2: *Sleep Delta Trend line*

## Risk Level Count

| | Risk level | | | |
|---|---|---|---|---|
| | High | Low | Medium | Not scor.. |
| Anxiety | 459 | 1,894 | 2,005 | 2,234 |
| Count of signalwell_.. | 175 | 724 | 761 | 840 |
| Mood | 493 | 2,690 | 2,538 | 2,870 |
| Movement Score | 9,447 | 39,207 | 41,795 | 45,949 |
| Risk Score | 849 | 874 | 2,109 | |
| Screen Minutes | 55,820 | 216,890 | 233,695 | 258,838 |
| Sleep 7-day baseline: | | 7 | 7 | 7 |
| Sleep Delta | | 1 | 0 | 0 |
| Sleep Hours | 1,151 | 5,294 | 5,573 | 6,107 |
| Stress | 630 | 1,927 | 2,294 | 2,484 |
| Wellbeing | 499 | 2,502 | 2,452 | 2,754 |

IMG 5.3: *Risk level count comparison*

**Result:** Advanced data analysis was successfully performed and an analytical report was generated using Tableau.

**Conclusion:** Tableau provides powerful analytical and reporting capabilities for extracting meaningful insights from research data.

## Experiment 6: Design and Development of Interactive Dashboards Using Tableau

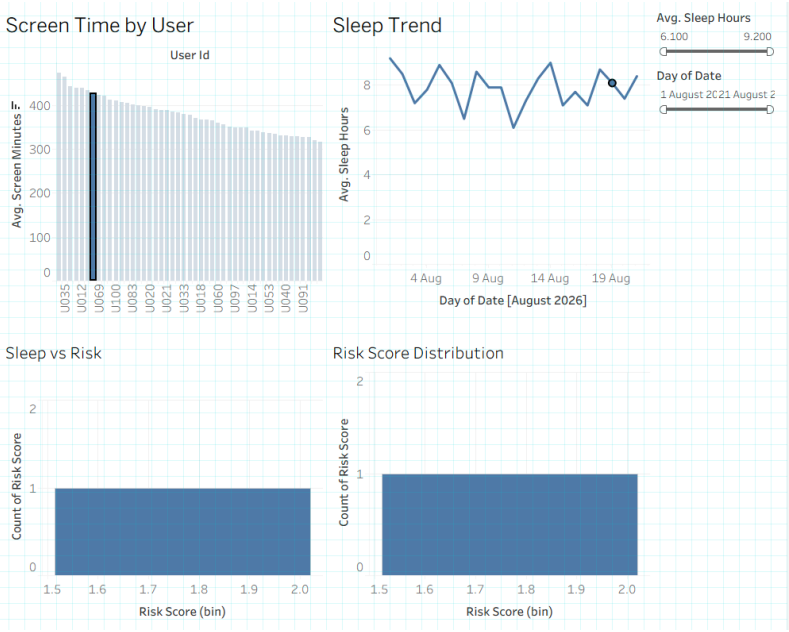
**Aim:** To design and develop an interactive dashboard using Tableau for effective visualization and analysis of research data.

**Tools Used:** Tableau

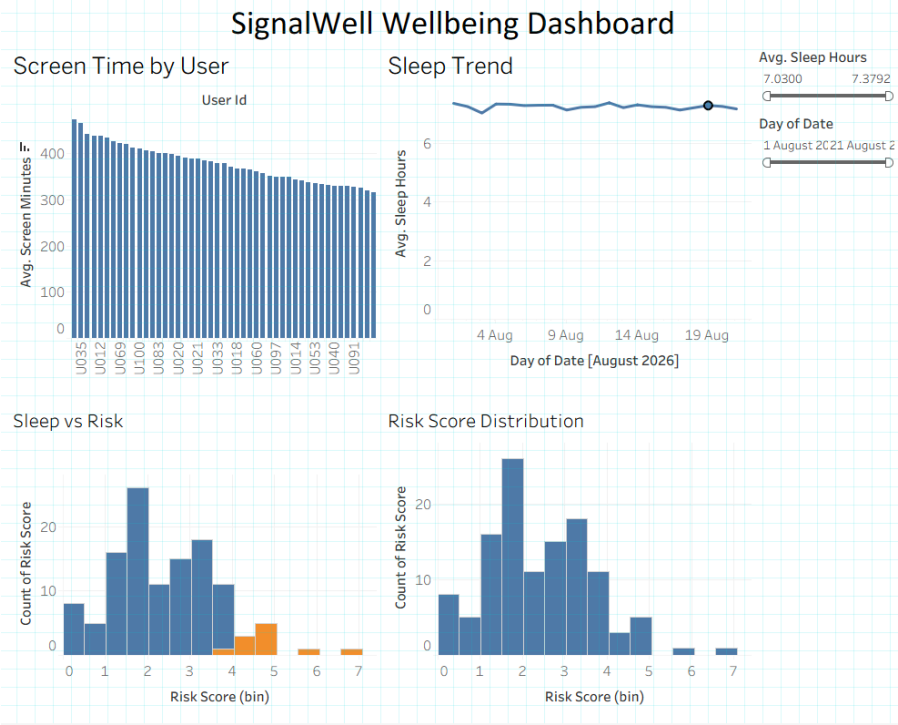
**Procedure:**

1. Open Tableau and import the required dataset.
2. Create the necessary charts and visualizations.
3. Arrange the visualizations within a dashboard.
4. Add filters, parameters and interactive controls.
5. Configure interactions between the dashboard components.
6. Format and finalize the dashboard for clear and effective presentation.

**Output:**




IMG 6.1: *SignalWell Dashboard Filtered to one user data*




IMG 6.2: *SignalWell interactive dashboard with parameters*

**Result:** An interactive dashboard was successfully designed and developed using Tableau.

**Conclusion:** Tableau enables the development of interactive dashboards that provide effective exploration and interpretation of research data.

## Experiment 7: Installation, Interface Exploration and Reference Management Using Mendeley

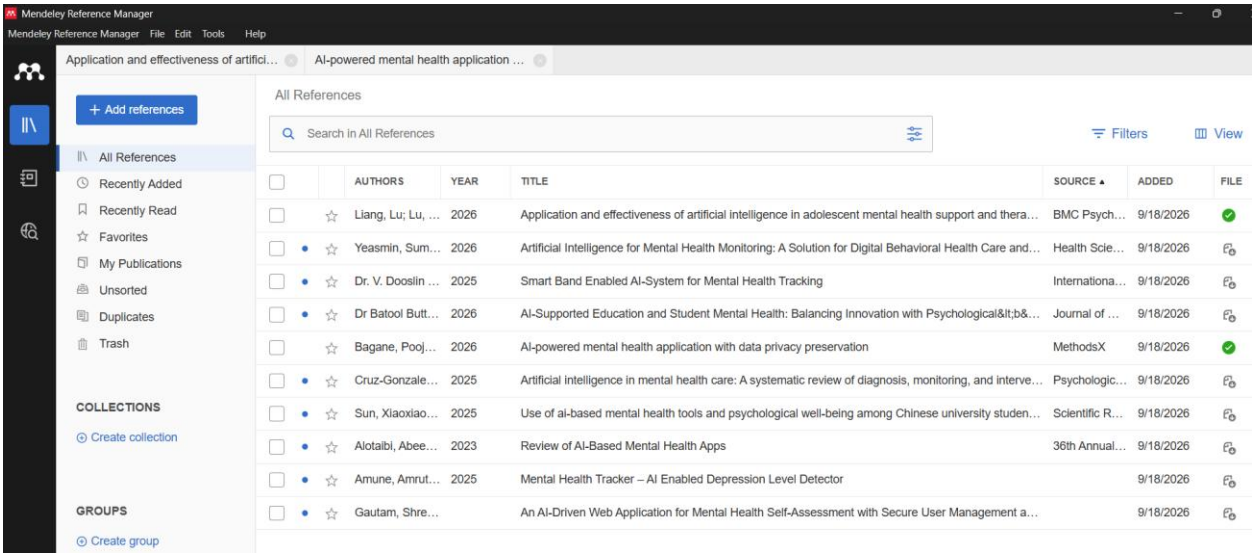
**Aim:** To install Mendeley, explore its interface and manage research references efficiently.

**Tools Used:** Mendeley

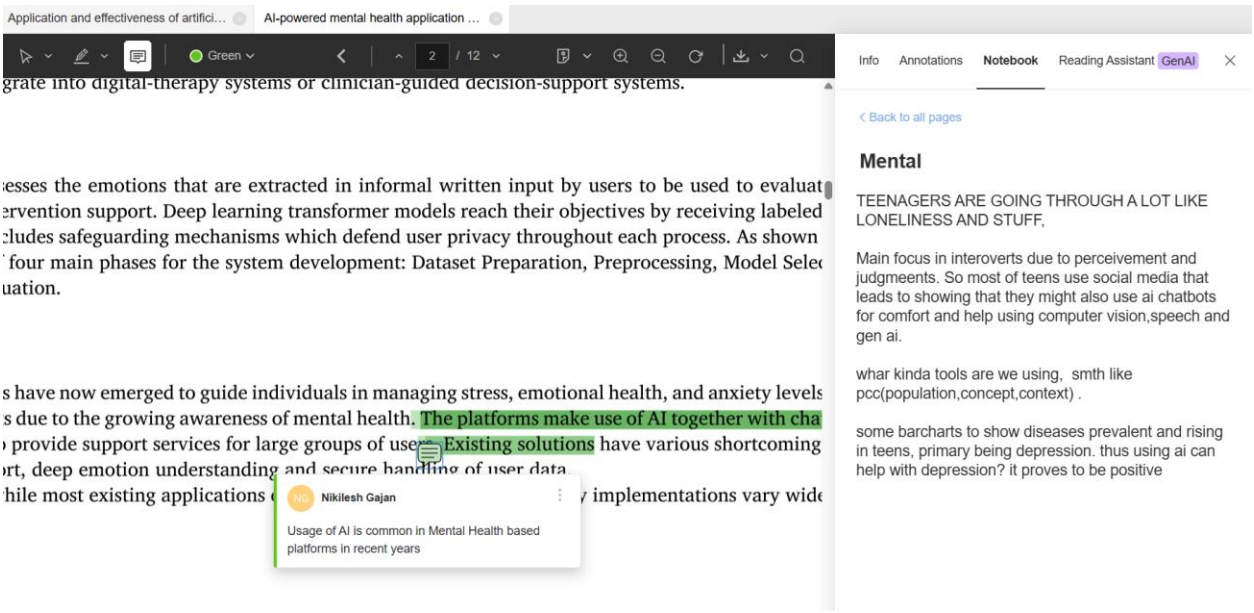
**Procedure:**

1. Download and install Mendeley.
2. Open Mendeley and explore its user interface.
3. Create a library for managing research references.
4. Add research articles and reference files to the library.
5. Organize references using folders, tags and relevant details.
6. Search, view and manage the stored references using Mendeley.

**Output:**




IMG 7.1: *References in Mendeley Reference library*




IMG 7.2: *Note making and Highlighting from the resources in Mendeley resource manager*

**Result:** Mendeley was successfully installed and used to organize and manage research references.

**Conclusion:** Mendeley provides an effective platform for storing, organizing and managing academic research references.

## Experiment 8: Metadata Verification, In-Text Citation and Bibliography Generation Using Mendeley

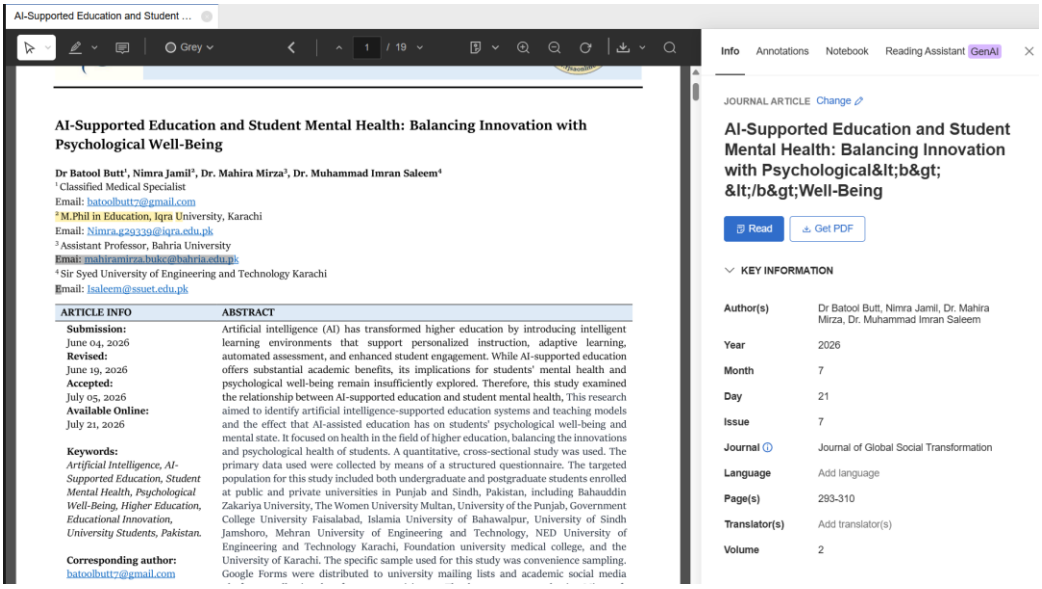
**Aim:** To verify research metadata and generate in-text citations and bibliographies using Mendeley.

**Tools Used:** Mendeley

**Procedure:**

1. Open Mendeley and select the required research articles.
2. Verify the metadata such as author, title, journal, year and publication details.
3. Correct or update any inaccurate metadata.
4. Insert the required references into a research document using Mendeley.
5. Generate in-text citations using an appropriate citation style.
6. Generate and format the bibliography using Mendeley.

**Output:**




IMG 8.1: *Metadata verification of a reference in Mendeley*

**Result:** Research metadata was successfully verified and in-text citations and bibliography were generated using Mendeley.

**Conclusion:** Mendeley simplifies metadata verification, citation management and bibliography generation for academic research.

## Experiment 9: Plagiarism Detection and Similarity Analysis Using Turnitin

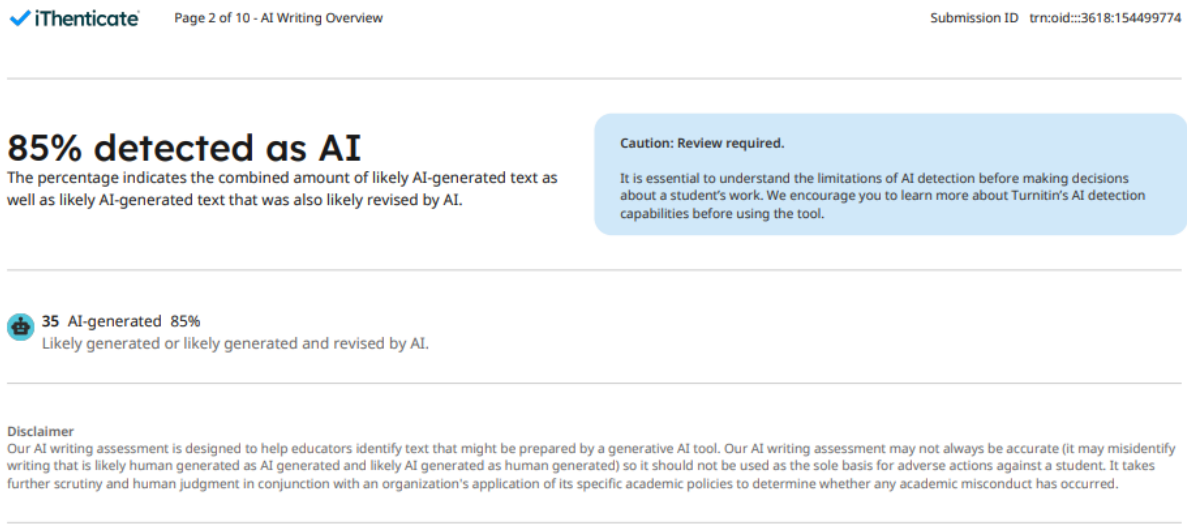
**Aim:** To detect plagiarism and analyze the similarity of a research document using Turnitin.

**Tools Used:** Turnitin

**Procedure:**

1. Open Turnitin and access the required assignment or submission area.
2. Prepare and upload the research document.
3. Submit the document for similarity checking.
4. Generate and open the similarity report.
5. Examine the highlighted similarities and matched sources.
6. Analyze the similarity percentage and identify sections requiring modification.

**Output:**




IMG 9.1: Plagiarism Check from Turnitin

**Result:** The research document was successfully analyzed for plagiarism and its similarity percentage was identified using Turnitin.

**Conclusion:** Turnitin provides an effective platform for detecting similarities and evaluating the originality of academic research documents.

## Experiment 10: Interpretation and Evaluation of Similarity Reports and Overleaf

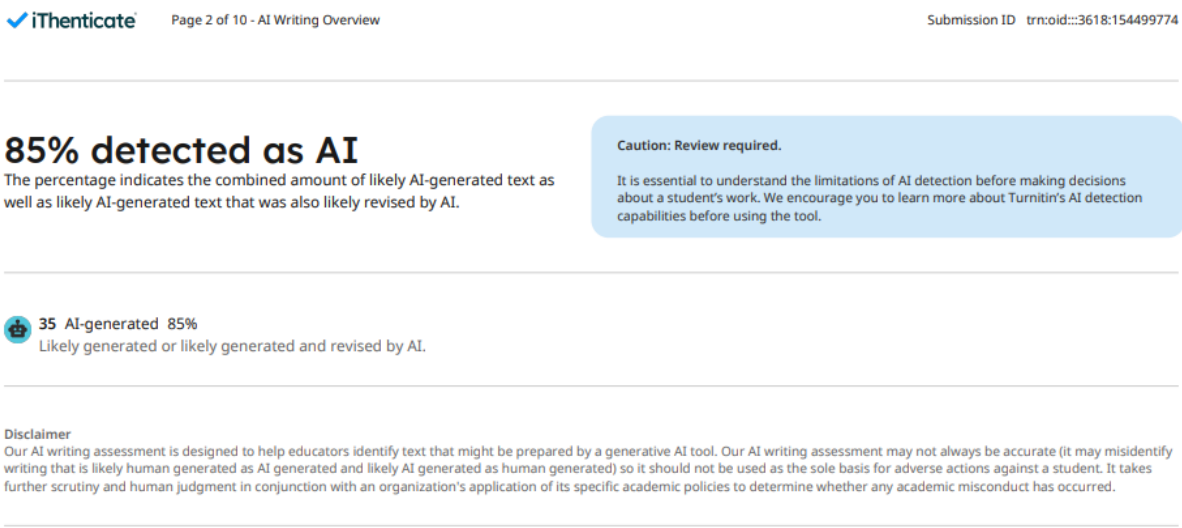
**Aim:** To interpret and evaluate similarity reports and prepare research documents using Overleaf.

**Tools Used:** Turnitin and Overleaf

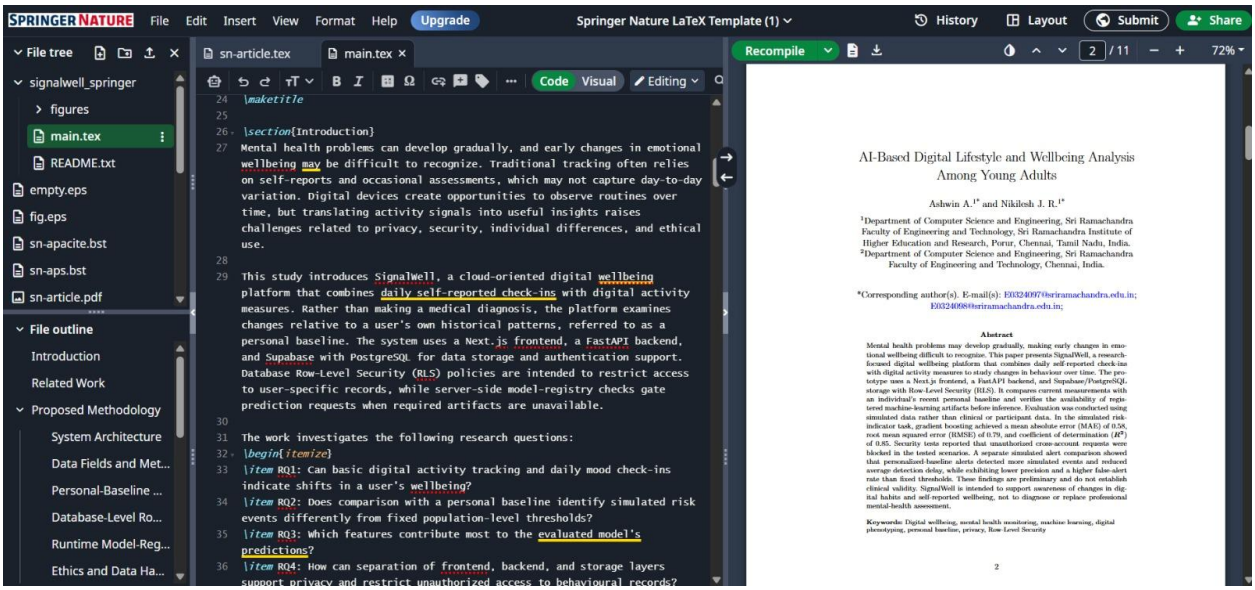
**Procedure:**

1. Open the generated Turnitin similarity report.
2. Examine the overall similarity percentage and matched sources.
3. Identify and evaluate the sections containing similarities.
4. Make necessary corrections and proper citations in the research document.
5. Open Overleaf and create a research document using an appropriate template.
6. Format the corrected research content and references using Overleaf.

**Output:**




*IMG 10.1: Turnitin similarity report*




IMG 10.2: *Research paper formatted in the Springer Nature template in Overleaf*

**Result:** The similarity report was successfully interpreted and evaluated, and the research document was formatted using Overleaf.

**Conclusion:** Turnitin helps evaluate the originality of research documents, while Overleaf provides an effective platform for preparing professionally formatted research documents.